ADC1: POS A1..A4 = GPIO1, 2, 3, 4   IS A1..A4 = GPIO5, 6, 7, 8   BUS_VS = GPIO9   V12_SNS = GPIO10
PWM R/L (active-low into 74HC14): A1 = 11/12 A2 = 13/14 A3 = 15/16 A4 = 17/18
I2C (BNO085 0x4A): SDA = 41, SCL = 42   IMU_INT = 20, IMU_RST = 19
SSR1 (PV) = GPIO21   SSR2 = GPIO47   ACT_EN = 39 (through E-stop loop)   ESTOP_SENSE = 40   LED_ERR = 46

IBT-2 IDC 2x5 (v3 pinout): 1 RPWM 2 LPWM 3 R_EN 4 L_EN (EN5 = E-stop gate) 5 R_IS 6 L_IS 7 VCC(+5V) 8 GND 9/10 n.c.
PV switch: two DELIXI CDG1-1DD/25A DC SSRs, outputs in series back-to-back (- to -).
Through-hole parts are hand-soldered (not in the JLC assembly BOM).

power

12 V input, 5 V / 3.3 V supplies, bus voltage monitor, SSR drivers

File: power.kicad_sch

mcu

ESP32-S3, IMU, E-stop

File: mcu.kicad_sch

actuators

Linear actuators: 4x IBT-2 (IDC 2x5), position & current sense

File: actuators.kicad_sch

Root sheet
Main controller - ESP32-S3
solbot

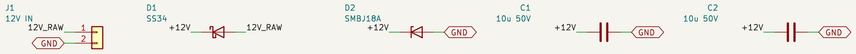
Sheet: /
File: solbot_main_v2.kicad_sch

Title: solbot_main v2 - compact ESP32-S3 controller

Size: A3	Date: 2026-10-15	**Rev: v2 rev B**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4

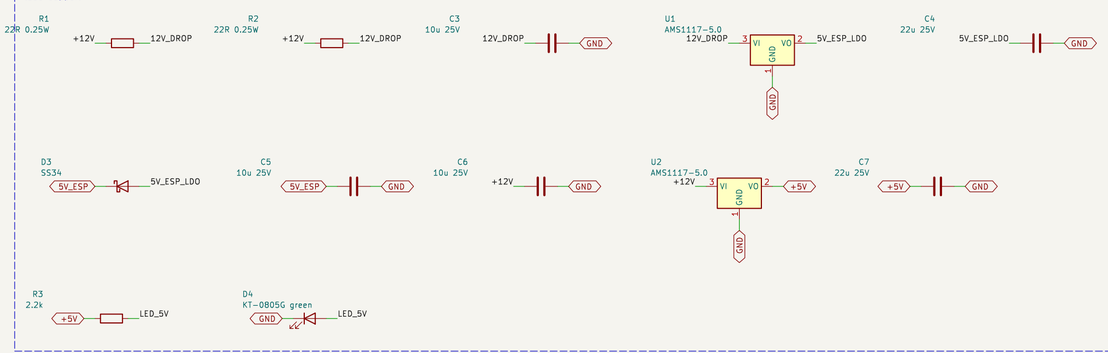
IN12

12 V input from the robot 12 V rail (fuse it externally, <= 2 A): SS34 reverse-polarity diode, SMBJ18A TVS, 2x10 uF.



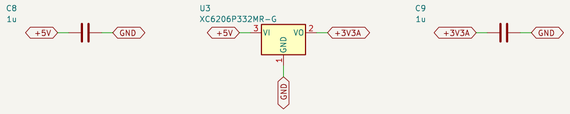
LDO5

Two AMS1117-5.0 linear regulators (Basic; v3 used 2x 7805). U_ESP feeds the DevKit through a 2x22R pre-drop (shares the heat: ~0.25 W in the resistors, ~0.7 W in the LDO at 150 mA) and SS34 (USB back-feed). U_LOG feeds +5V: IBT-2 logic, 74HC14, IMU, 3.3 V LDO. Give both SOT-223 tabs copper.



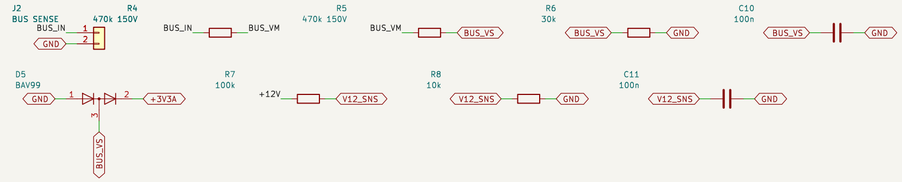
LDO3

XC6206P332MR 3.3 V (200 mA): I2C pull-ups, pot excitation, clamps.



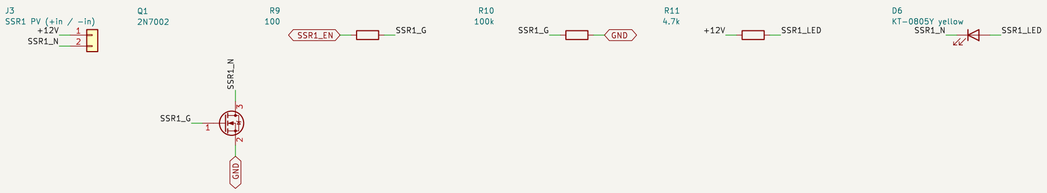
VMON

Main battery bus monitor: (470k+470k+30k)/30k = 32.3 -> 2.94 V @ 95 V (10S..20S) to GPIO9 (ADC1); 12 V rail 110k/10k -> GPIO10. Bus GND must be common with the 12 V GND (non-isolated DC-DC).



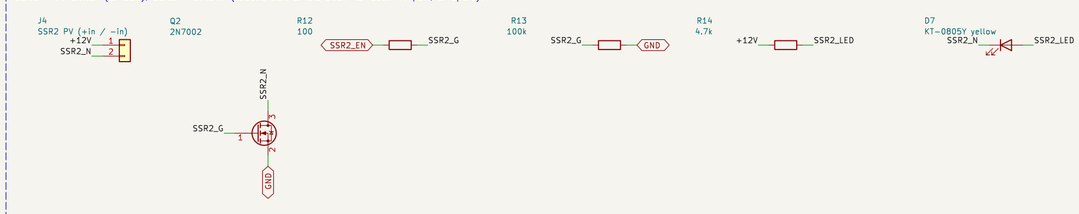
SSR1

SSR1 driver: DC SSR input (3-32 V, ~10 mA) between +12V and a low-side 2N7002. LED = SSR on.
SSR1 = PV switch (GPIO21), SSR2 = GPIO47 (second SSR of the back-to-back PV pair, or spare).



SSR2

SSR2 driver: DC SSR input (3-32 V, ~10 mA) between +12V and a low-side 2N7002. LED = SSR on.
SSR1 = PV switch (GPIO21), SSR2 = GPIO47 (second SSR of the back-to-back PV pair, or spare).



Sheet: power
Main controller — ESP32-S3
solbot
Sheet: /power/
File: power.kicad_sch
Title: 12 V input, 5 V / 3.3 V supplies, bus voltage monitor, SSR drivers
Size: A2 | Date: 2026-10-15 | **Rev: v2 rev B**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 2/4

ESP

ESP32-S3-DevKitC-1 in 2x22 sockets (hand-soldered). ADC1 GPIO1-10 all used (pots, IS currents, bus, 12 V). GPIO19/20 used -> program via the "UART" USB port. GPIO35-37 free (octal PSRAM on R8 variants), GPIO38/48 = on-board RGB LED.

U4
ESP32-S3-DevKitC-1

	1	3V3	GND	23	GND
	2	3V3	TX/GPIO43	24	
	3	RST	RX/GPIO44	25	
A4_POS	4	GPIO4	GPIO1	26	A1_POS
A1_ISA	5	GPIO5	GPIO2	27	A2_POS
A2_ISA	6	GPIO6	GPIO42	28	SCL
A3_ISA	7	GPIO7	GPIO41	29	SDA
A3_RPWM	8	GPIO15	GPIO40	30	ESTOP_SENSE
A3_LPWM	9	GPIO16	GPIO39	31	ACT_EN
A4_RPWM	10	GPIO17	GPIO38	32	
A4_LPWM	11	GPIO18	GPIO37	33	
A4_ISA	12	GPIO8	GPIO36	34	
A3_POS	13	GPIO3	GPIO35	35	
LED_ERR	14	GPIO46	GPIO0	36	
BUS_VS	15	GPIO9	GPIO45	37	
V12_SNS	16	GPIO10	GPIO48	38	
A1_RPWM	17	GPIO11	GPIO47	39	SSR2_EN
A1_LPWM	18	GPIO12	GPIO21	40	SSR1_EN
A2_RPWM	19	GPIO13	GPIO20	41	IMU_INT
A2_LPWM	20	GPIO14	GPIO19	42	IMU_RST
5V_ESP	21	5V	GND	43	GND
GND	22	GND	GND	44	GND

R15
1k
LED_ERR — LED_ERR_A

D8
KT-0603R red
GND — LED_ERR_A

R16
4.7k
+3V3A — SDA

R17
4.7k
+3V3A — SCL

ESTOP

Hardware E-stop -> IBT-2 enable: ACT_EN (GPIO39) runs through the NC E-stop loop to Q_A; Q_B inverts again, so EN5 (IDC pins 3/4 = R_EN/L_EN) is high only while the loop is closed AND the ESP drives ACT_EN. ESP reset / loop open -> EN5 low -> bridges off. Fit a wire bridge if no E-stop is used.

J5
E-STOP (NC loop)
ESTOP_SRC 1
ESTOP_RET 2

R18
470
ACT_EN — ESTOP_SRC

R19
1k
ESTOP_RET — ACT_GATE

R20
100k
ACT_GATE — GND

C12
100n
ACT_GATE — GND

R21
10k
ESTOP_RET — ESTOP_SENSE

Q3
2N7002
EN_N -> 3
ACT_GATE -> 1
2 -> GND

R22
10k
+5V — EN_N

Q4
2N7002
ACT_EN5 -> 3
EN_N -> 1
2 -> GND

R23
2.2k
+5V — ACT_EN5

IMU

BNO085 (Adafruit 4754) on cable: 5 V, I2C @0x4A, INT -> GPIO20, RST <- GPIO19.

J6
IMU BNO085

+5V	1
GND	2
SCL	3
SDA	4
IMU_INT	5
IMU_RST	6

R24
10k
+3V3A — IMU_INT

C13
10u 25V
+5V — GND

MECH

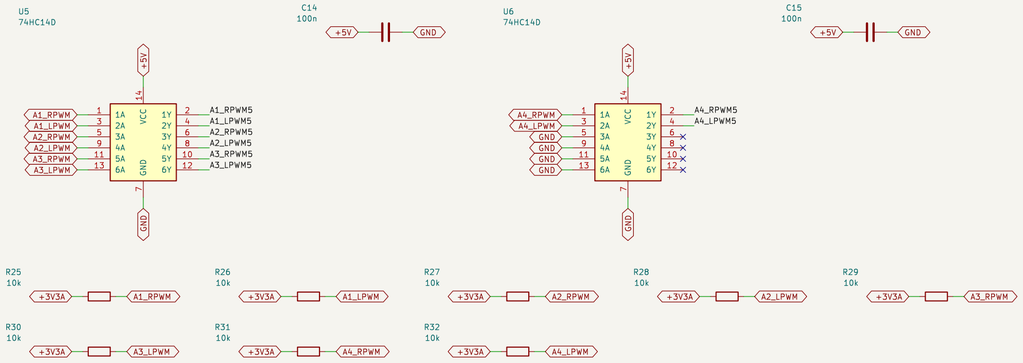
Mounting holes M3.

H1 M3 H2 M3 H3 M3 H4 M3

Sheet: mcu
Main controller — ESP32-S3
solbot
Sheet: /mcu/
File: mcu.kicad_sch
Title: ESP32-S3, IMU, E-stop
Size: A3 | Date: 2026-10-15 | **Rev: v2 rev B**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 3/4

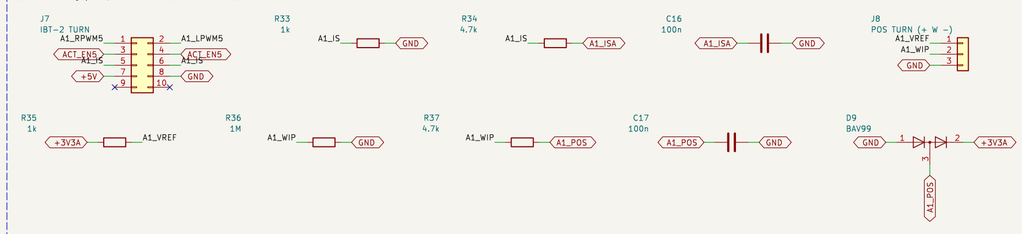
BUF

2x 74HC14 (Basic) shift 3.3 V -> 5 V and INVERT; firmware drives the PWM active-low, 10k pull-ups on the ESP side keep the IBT-2 inputs low while the ESP boots (the EN gate keeps the bridges disabled anyway). Unused inputs tied to GND.



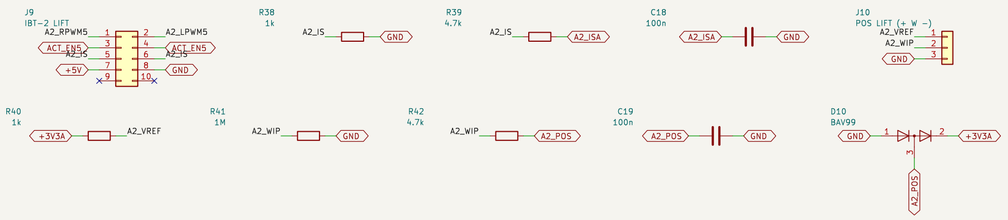
A1

Actuator 1 (TURN). IDC 2x5 (v3 pinout, flat cable 1:1 to the IBT-2 header): 1 RPWM, 2 LPWM, 3 R_EN, 4 L_EN (both = EN5 gate), 5 R_IS, 6 L_IS, 7 VCC, 8 GND, 9/10 n.c. IS: 1k (DNP if the module has one) -> 1.18 V @ 10 A. Pot: 3V3A via 1k (keeps the wiper below ~3.0 V, inside the ESP ADC range), wiper -> 4.7k/100n -> ADC1.



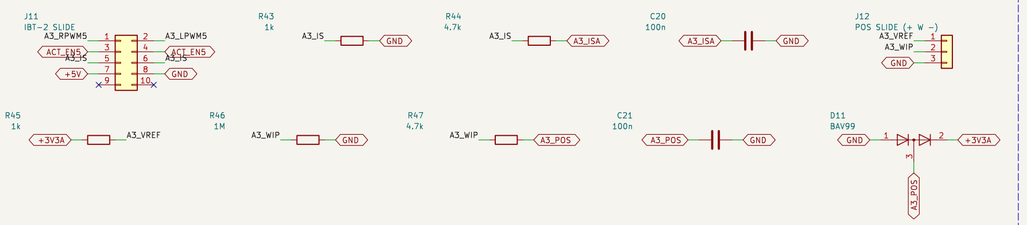
A2

Actuator 2 (LIFT). IDC 2x5 (v3 pinout, flat cable 1:1 to the IBT-2 header): 1 RPWM, 2 LPWM, 3 R_EN, 4 L_EN (both = EN5 gate), 5 R_IS, 6 L_IS, 7 VCC, 8 GND, 9/10 n.c. IS: 1k (DNP if the module has one) -> 1.18 V @ 10 A. Pot: 3V3A via 1k (keeps the wiper below ~3.0 V, inside the ESP ADC range), wiper -> 4.7k/100n -> ADC1.



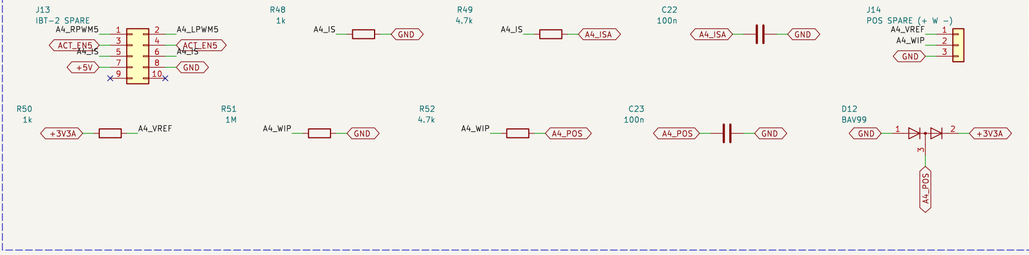
A3

Actuator 3 (SLIDE). IDC 2x5 (v3 pinout, flat cable 1:1 to the IBT-2 header): 1 RPWM, 2 LPWM, 3 R_EN, 4 L_EN (both = EN5 gate), 5 R_IS, 6 L_IS, 7 VCC, 8 GND, 9/10 n.c. IS: 1k (DNP if the module has one) -> 1.18 V @ 10 A. Pot: 3V3A via 1k (keeps the wiper below ~3.0 V, inside the ESP ADC range), wiper -> 4.7k/100n -> ADC1.



A4

Actuator 4 (SPARE). IDC 2x5 (v3 pinout, flat cable 1:1 to the IBT-2 header): 1 RPWM, 2 LPWM, 3 R_EN, 4 L_EN (both = EN5 gate), 5 R_IS, 6 L_IS, 7 VCC, 8 GND, 9/10 n.c. IS: 1k (DNP if the module has one) -> 1.18 V @ 10 A. Pot: 3V3A via 1k (keeps the wiper below ~3.0 V, inside the ESP ADC range), wiper -> 4.7k/100n -> ADC1.



Sheet: actuators
Main controller — ESP32-S3
solbot

Sheet: /actuators/
File: actuators.kicad_sch

Title: Linear actuators: 4x IBT-2 (IDC 2x5), position & current sense

Size: A2	Date: 2026-10-15	**Rev: v2 rev B**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/4